THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

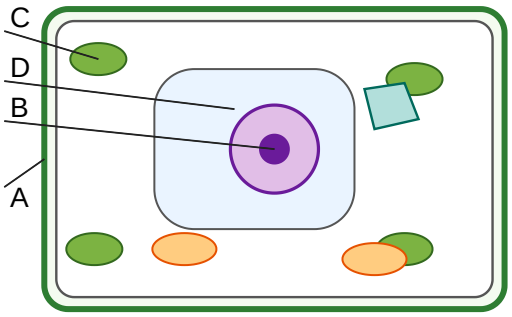
ANSWER KEY & MARKING SCHEME — SCIENCE · CLASS 9 · SET 05

Max. Marks: 80 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — BIOLOGY — 16 questions · 32 marks

Q.1–7 MCQs (1 mark each); Q.8–9 Assertion–Reason. For Assertion–Reason: (a) Both A and R are true, and R is the correct explanation of A. (b) Both A and R are true, but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

1. In the plant cell shown, the structure that gives the cell a definite shape and protects it is labelled:



(a) A (b) B (c) C (d) D

Ans: (a) The cell wall (A) gives shape and rigidity.

2. Which of the following is NOT a part of the cell theory?

(a) All organisms are made of cells

(b) The cell is the structural and functional unit of life

(c) All cells arise from pre-existing cells

(d) All cells contain chloroplasts

Ans: (d) Only plant cells contain chloroplasts.

3. A gardener cuts the tip of a rose stem so that the plant becomes bushy. Further growth of that stem in length stops because the following tissue is removed:

(a) apical meristem (b) lateral meristem

(c) intercalary meristem (d) collenchyma

Ans: (a) Apical meristem at the tip is responsible for growth in length.

4. Which feature distinguishes cardiac muscle from skeletal muscle?

(a) Presence of striations (b) Involuntary action and branched fibres

(c) Presence of nuclei (d) Cylindrical cells

Ans: (b) Cardiac muscle is involuntary with branched fibres; skeletal is voluntary and unbranched.

5. The insolation at a place is 0.6 kW m^{-2}. The solar energy received by 2 m^2 of area in 30 minutes is:

(a) 0.6 kWh (b) 1.2 kWh (c) 0.3 kWh (d) 2.4 kWh

Ans: (a) 0.6 × 2 × 0.5 = 0.6 kWh.

6. Which of the following pairs is correctly matched?

(a) Monera – nucleated cells (b) Fungi – contain chlorophyll

(c) Protista – unicellular eukaryotes (d) Plantae – heterotrophs

Ans: (c) Protista are unicellular eukaryotes.

7. In human beings, fertilisation normally takes place in the:

(a) uterus (b) oviduct (fallopian tube)
(c) ovary (d) vagina

> **Ans: (b)** The sperm fertilises the egg in the oviduct.

8. **Assertion (A):** Plant cells do not burst when placed in pure water.
Reason (R): The cell wall of a plant cell resists the swelling of the cell.

> **Ans: (a)** R is the reason why plant cells withstand endosmosis.

9. **Assertion (A):** Burning fossil fuels adds carbon that was locked underground to the atmosphere.
Reason (R): The carbon cycle is not influenced by human activities.

> **Ans: (c)** A is true; R is false – human activity has disturbed the cycle.

10. Differentiate between voluntary and involuntary muscles, with one example of each.

> Voluntary – under conscious control (biceps); involuntary – not under control (heart/intestine). [1 + 1]

O R

Why are cartilage and bone called connective tissues? State one function of each.

> Both have cells embedded in a matrix and connect/support other tissues [1]; cartilage – smooth surfaces at joints; bone – support/protection. [1]

11. Describe the structure of the nucleus. State any two functions.

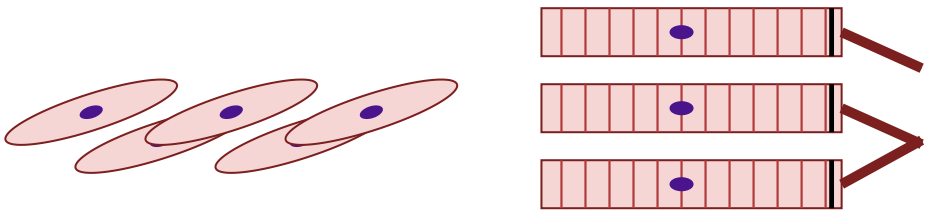
> Double-layered nuclear membrane with pores, nucleoplasm, chromatin (chromosomes), nucleolus [1½]; controls cell activities; carries hereditary information (DNA). [1½]

O R

Name the cell organelle that (i) packages and dispatches materials, (ii) digests worn-out parts and (iii) synthesises lipids. Mention where each is present.

> (i) Golgi apparatus (ii) Lysosome (iii) Smooth ER [3 × 1]

12. The figure shows two types of muscle fibres P and Q. Identify them, name one organ where each is found and state one difference between them.



P **Q**

> P – smooth muscle (stomach/intestine wall); Q – cardiac muscle (heart) [1½]; smooth – unstriated, spindle-shaped, single nucleus; cardiac – striated, branched. [1½]

13. Fill in the blanks:

Method of reproduction	Organism
Budding	(a) _____
Fragmentation	(b) _____
Spore formation	(c) _____

> (a) Hydra / yeast (b) Spirogyra (c) Rhizopus / mushroom / fern [1 each]

14. Name three greenhouse gases and write one source of each.

CO_2 – burning fossil fuels/respiration; CH_4 – paddy fields/livestock/landfills; N_2O – fertilisers; water vapour; CFCs. [Any three × 1]

15. A solar farm in Rajasthan has solar panels covering 50 m^2. The average insolation at the site is 0.8 kW m^{-2} and the panels receive sunlight for 6 hours a day.

(a) Calculate the solar energy received by the panels in a day.

0.8 × 50 × 6 = 240 kWh. [1]

(b) Why is the insolation received at the Earth's surface less than the solar constant?

Part of the radiation is absorbed and reflected by the atmosphere (clouds, gases, dust). [1]

(c) Suggest two ways of reducing the carbon footprint of a household.

Use renewable (solar) energy, public transport/cycling, save electricity, plant trees, reduce waste. [Any two × 1]

OR

(c) How do greenhouse gases keep the Earth warm? Why is their excess harmful?

They trap outgoing infrared radiation [1]; excess traps more heat → global warming and climate change. [1]

16. Differentiate between xylem and phloem on any five points.

Function; type of cells (dead/living); cell types (vessels, tracheids/ sieve tubes, companion cells); direction of transport; lignified walls; materials. [Any five × 1]

OR

Describe any four types of epithelial tissue, mentioning one location of each.

Squamous (alveoli, lining of mouth) [1¼]; cuboidal (kidney tubules, glands) [1¼]; columnar (intestine lining) [1¼]; ciliated columnar (respiratory tract) [1¼].

SECTION B — CHEMISTRY — 11 questions · 25 marks

Q.17–19 MCQs; Q.20 Assertion–Reason (options as in Section A).

17. Which of the following mixtures will show the Tyndall effect?

(a) Fog (b) Sugar solution (c) Vinegar (d) Brass

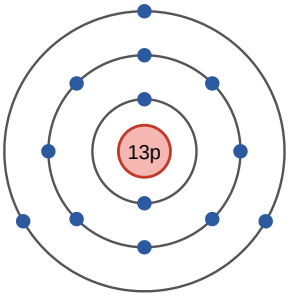
Ans: (a) Fog is a colloid (liquid in gas) – it scatters light.

18. 5.6 g of iron combines with 3.2 g of sulphur to form 8.8 g of iron sulphide. The mass of iron sulphide formed when 11.2 g of iron combines completely with sulphur is:

(a) 8.8 g (b) 11.2 g (c) 17.6 g (d) 22.4 g

Ans: (c) Doubling the iron needs doubling the sulphur (6.4 g): 17.6 g (law of constant proportions).

19. The atom shown has 13 protons. The charge on the ion it normally forms is:



(a) 3− (b) 3+ (c) 1+ (d) 2+

Ans: (b) Configuration 2, 8, 3 – it loses 3 electrons to form Al^{3+}.

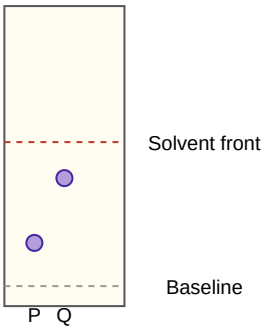
20. **Assertion (A):** Paper chromatography can be used to separate the components of ink.
Reason (R): Different components of ink have different solubilities in the solvent and adsorb differently on the paper.

Ans: (a) R explains the principle of the separation in A.

21. The radius of an atom is about 1×10^{-10} m and that of its nucleus is about 1×10^{-15} m. How many times larger is the atom than the nucleus? Express the answer in scientific notation.

Ratio = $10^{-10}/10^{-15}$ [1] = 1×10^{5} times [1]

22. The chromatogram of an ink is shown. The solvent front is 80 mm from the baseline; spot P is 24 mm and spot Q is 60 mm from the baseline. Calculate the R_f values of P and Q.



R_f = distance of spot ÷ distance of solvent front. P = 24/80 = 0.30 [1]; Q = 60/80 = 0.75 [1]

23. Write the chemical formula of (i) calcium hydroxide and (ii) ammonium sulphate.

(i) $Ca(OH)_2$ [1] (ii) $(NH_4)_2SO_4$ [1]

24. 5 g of a solute is dissolved in water to make 250 mL of solution. (i) Calculate the mass by volume percentage (% m/v). (ii) How much solute is present in 1 litre of this solution?

(i) 5/250 × 100 = 2 % m/v [1½] (ii) 2 g per 100 mL → 20 g per litre [1½]

25. State any two postulates of Dalton's atomic theory and one limitation of the theory.

Matter is made of indivisible atoms; atoms of an element are identical; atoms combine in fixed ratios; atoms cannot be created or destroyed. [2] Limitation – atoms are divisible (electrons, protons, neutrons) / isotopes exist. [1]

OR

State the law of conservation of mass. 12 g of carbon burns in 32 g of oxygen. Find the mass of carbon dioxide formed.

In a chemical reaction mass is neither created nor destroyed [1]; mass of CO_2 = 12 + 32 [1] = 44 g [1]

26. The table gives the number of sub-atomic particles in four species A, B, C and D.

Species	Protons	Neutrons	Electrons
A	6	6	6
B	6	8	6
C	7	7	7
D	11	12	10

(a) Which two species are isotopes of the same element?

A and B (same protons, different neutrons). [1]

(b) What is the mass number of species D?

> 11 + 12 = 23. [1]

(c) What is the charge on D? Justify.

> +1 [1]; protons (11) exceed electrons (10) by one. [1]

OR

(c) Are B and C isobars? Give a reason.

> Yes [1]; B: 6 + 8 = 14, C: 7 + 7 = 14 – same mass number, different atomic numbers. [1]

27. Distinguish between electron, proton and neutron on the basis of (i) charge, (ii) relative mass, (iii) location, (iv) symbol and (v) discoverer.

> Electron: −1, negligible (1/1836 u), outside nucleus, e^-, J. J. Thomson; Proton: +1, 1 u, nucleus, p^+, Goldstein (canal rays)/Rutherford; Neutron: 0, 1 u, nucleus, n, Chadwick. [5 × 1]

OR

Describe Rutherford's α-particle scattering experiment. State three observations and two conclusions.

> Thin gold foil bombarded with α-particles [1]. Observations – most passed straight through; some deflected through small angles; very few (≈1 in 20 000) rebounded. [1½] Conclusions – most of atom is empty space; positive charge and mass concentrated in a tiny nucleus. [2½]

SECTION C — PHYSICS — 12 questions · 23 marks

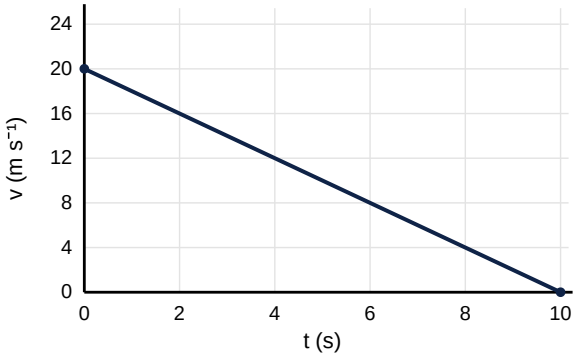
Q.28–33 MCQs; Q.34 Assertion–Reason (options as in Section A).

28. A car goes from A to B, 120 km apart, at 60 km h^{-1} and returns at 40 km h^{-1}. Its average speed for the round trip is:

(a) 50 km h^{-1} (b) 48 km h^{-1} (c) 52 km h^{-1} (d) 100 km h^{-1}

> **Ans: (b)** Total distance 240 km; time = 2 + 3 = 5 h; speed = 48 km h^{-1}.

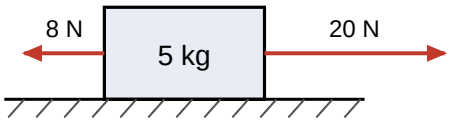
29. The velocity–time graph of a body is shown. The magnitude of its acceleration is:



(a) 0.5 m s^{-2} (b) 2 m s^{-2} (c) 10 m s^{-2} (d) 20 m s^{-2}

> **Ans: (b)** Slope = (0 − 20)/10 = −2 → magnitude 2 m s^{-2}.

30. A 5 kg block is pulled by a horizontal force of 20 N against a friction force of 8 N, as shown. Its acceleration is:



(a) 4 m s^{-2} (b) 2.4 m s^{-2} (c) 1.6 m s^{-2} (d) 5.6 m s^{-2}

Ans: (b) Net force = 12 N; a = 12/5 = 2.4 m s⁻².

31. A 0.2 kg ball strikes a wall at 10 m s^{-1} and rebounds at 10 m s^{-1} in the opposite direction. The magnitude of the change in its momentum is:

(a) 0 (b) 2 kg m s^{-1} (c) 4 kg m s^{-1} (d) 20 kg m s^{-1}

Ans: (c) Δp = 0.2 × (10 − (−10)) = 4 kg m s⁻¹.

32. A person lifts a 10 kg bag through 2 m in 4 s. The power developed is (g = 10 m s^{-2}):

(a) 20 W (b) 50 W (c) 80 W (d) 200 W

Ans: (b) P = mgh/t = 10×10×2/4 = 50 W.

33. In a longitudinal sound wave in air, the regions of high pressure are called:

(a) rarefactions (b) crests (c) compressions (d) troughs

Ans: (c) Compressions – regions of high pressure.

34. **Assertion (A):** A person is able to walk forward on a rough road.
Reason (R): Friction between the feet and the ground provides the forward push.

Ans: (a) R explains A – friction opposes the backward push of the foot.

35. A 50 kg man climbs 20 stairs, each 20 cm high. Calculate the work done against gravity and the gain in his potential energy (g = 10 m s^{-2}).

Height = 20 × 0.2 = 4 m [½]; W = mgh = 50×10×4 = 2000 J [1]; gain in PE = 2000 J [½]

36. Define amplitude of a sound wave. How does it affect the loudness of sound?

Maximum displacement of particles from the mean position [1]; larger amplitude → louder sound. [1]

OR

What is reverberation? State one way to reduce it in an auditorium.

Persistence of sound due to repeated reflections [1]; use curtains/carpets/sound-absorbing panels. [1]

37. A body falls freely from rest (g = 10 m s^{-2}). Find (i) its velocity after 2.5 s, (ii) the distance fallen in 2.5 s and (iii) the time taken to fall through 125 m.

(i) v = gt = 25 m s⁻¹ [1] (ii) h = ½gt² = 31.25 m [1] (iii) t = √(2h/g) = √25 = 5 s [1]

38. A swimmer pushes the water backwards with her hands and moves forward. A student also released an inflated balloon with its mouth open and noticed that it flew away in the opposite direction to the escaping air.

(a) Which law of motion explains both observations?

Newton's third law. [1]

(b) What is the reaction force when the swimmer pushes the water backwards?

The water pushes the swimmer forward with an equal force. [1]

(c) If action and reaction are equal and opposite, why do they not cancel each other?

They act on different bodies [1]; so they cannot be added to get zero net force on one body. [1]

OR

(c) A gun of mass 2 kg fires a bullet of mass 20 g at 100 m s^{-1}. Find the recoil velocity of the gun.

0.02 × 100 = 2 × v [1]; v = 1 m s⁻¹ opposite to the bullet. [1]

39. A source of frequency 400 Hz produces sound in air (speed 340 m s^{-1}). Find (i) the wavelength, (ii) the time period, (iii) the number of waves produced in 5 s, (iv) the wavelength in water where the speed is 1500 m s^{-1} and (v) the distance of a cliff if an echo is heard 3 s after a shout.

> (i) λ = 340/400 = 0.85 m [1] (ii) T = 1/400 = 0.0025 s [1] (iii) 400 × 5 = 2000 [1] (iv) 1500/400 = 3.75 m [1] (v) d = 340 × 3/2 = 510 m [1]

O R

A 4 kg body is thrown vertically upwards with a velocity of 20 m s^{-1} (g = 10 m s^{-2}). Find (i) its initial kinetic energy, (ii) the maximum height, (iii) its potential energy at the maximum height, (iv) its kinetic energy at a height of 10 m and (v) its speed at that height.

> (i) ½×4×400 = 800 J [1] (ii) h = $v^2/2g$ = 20 m [1] (iii) 800 J [1] (iv) PE at 10 m = 4×10×10 = 400 J, KE = 800 − 400 = 400 J [1] (v) ½×4×v^2 = 400 → v = √200 ≈ 14.1 m s^{-1} [1]